Supplementary Material

Machine-learning analysis of blood metals and metabolic syndrome in NHANES 2007–2018

Table S1. Predictor dictionary. Each NHANES item code is given with a plain description of what the item records, its coding in this analysis, and the number of model columns it contributes after encoding (total 33). Categorical variables with more than two levels were one-hot encoded with the stated reference category omitted.

Domain	NHANES code	What the item records	Coding	Cols
Blood metals	LBXBPB	Blood lead concentration	Continuous, µg/dL	1
	LBXBCD	Blood cadmium concentration	Continuous, µg/L	1
	LBXTHG	Blood total mercury concentration	Continuous, µg/L	1
	LBXBSE	Blood selenium concentration	Continuous, µg/L	1
	LBXBMN	Blood manganese concentration	Continuous, µg/L	1
Demographic and socioeconomic	RIDAGEYR	Age at screening	Continuous, years	1
	RIAGENDR	Sex	Binary (female = 1)	1
	RIDRETH1	Race/Hispanic origin	One-hot; Mexican American = reference	5
	DMDEDUC2	Highest education level completed (adults)	Ordinal 1–5 (<9th grade → college graduate)	1
	DMDMARTL	Marital status	One-hot; married = reference	6
	DMDHHSIZ	Total number of people in the household	Continuous, count	1
	INDFMPIR	Ratio of family income to the federal poverty threshold	Continuous, 0–5	1
	RIDEXMON	Six-month period in which the examination took place	Binary (May–October = 1)	1
Anthropometric	BMXBMI	Body mass index from measured height and weight	Continuous, kg/m²	1
Lifestyle	SMQ020	Smoked at least 100 cigarettes in life	Binary (yes = 1)	1
	SMQ040	Current cigarette smoking frequency	Ordinal, recoded 0 = not at all, 1 = some days, 2 = every day	1
	ALQ101 / ALQ111	Alcohol gatekeeper item, harmonized across cycles: "had at least 12 alcoholic drinks in any one year" (2007–2016) and "ever had a drink of any kind of alcohol" (2017–2018)	Binary (ever = 1)	1
	ALQ130	Average number of alcoholic drinks per day on drinking days, past 12 months	Continuous, drinks/day	1
	DBQ700	Self-rated overall diet quality	Ordinal 1–5 (excellent → poor)	1
	PAQ605	Vigorous-intensity activity as part of work	Binary (yes = 1)	1

Domain	NHANES code	What the item records	Coding	Cols
	PAQ650	Vigorous-intensity recreational activity	Binary (yes = 1)	1
	PAQ665	Moderate-intensity recreational activity	Binary (yes = 1)	1
	PAD645	Minutes spent walking or bicycling for transportation on a typical day (gated on PAQ635, "walk or use a bicycle for transportation")	Continuous, minutes/day	1
	BPXPLS	Resting pulse, 60-second count	Continuous, beats/min	1

Survey modules used, all merged on the respondent sequence number (SEQN): demographics (DEMO), body measures (BMX), blood pressure (BPX), triglycerides (TRIGLY), HDL cholesterol (HDL), plasma fasting glucose (GLU), blood lead–cadmium–mercury–selenium–manganese (PBCD), alcohol use (ALQ), diet behavior (DBQ), smoking (SMQ), diabetes (DIQ), physical activity (PAQ), and blood pressure and cholesterol (BPQ). Hip circumference (BMXHIP) and physician-diagnosed diabetes (DIQ010) were screened but not retained in the primary predictor set.

Table S2. Metabolic syndrome component definitions. A component is scored positive when either the measured criterion or the drug-treatment criterion is met, following the harmonized joint interim statement [5] and the earlier AHA/NHLBI thresholds [6]. Participants meeting at least three of five components were classified as cases.

Component	Measured criterion	NHANES variable	Drug-treatment criterion
Abdominal obesity	Waist circumference ≥ 102 cm (men) or ≥ 88 cm (women)	BMXWAIST	Not applicable
Elevated triglycerides	Triglycerides ≥ 150 mg/dL	LBXTR	BPQ100D (currently taking prescribed medicine for cholesterol)
Reduced HDL cholesterol	HDL-C < 40 mg/dL (men) or < 50 mg/dL (women)	LBDHDD	BPQ100D (currently taking prescribed medicine for cholesterol)
Elevated blood pressure	Systolic ≥ 130 mmHg or diastolic ≥ 85 mmHg	BPXSY1, BPXDI1	BPQ050A (currently taking prescribed medicine for hypertension)
Elevated fasting glucose	Fasting plasma glucose ≥ 100 mg/dL	LBXGLU	DIQ050 (currently taking insulin); DIQ070 (taking pills to lower blood sugar)

HDL-C, high-density lipoprotein cholesterol. NHANES carries only one cholesterol-medication item in the cycles used here and does not distinguish HDL-specific from triglyceride-specific treatment, so BPQ100D is applied to both lipid components. Component measurements and medication items were used solely to construct the outcome and were excluded from the predictor set.

Table S3. Hyperparameter search space, selected configuration and cross-validated discrimination for each algorithm. Configurations were chosen by randomized search (capped at 12 configurations; smaller grids searched exhaustively) with 10-fold stratified cross-validation on the training partition, optimizing AUROC. *These values are tuning-selection statistics and are optimistically biased; test-partition performance is reported in the main text.*

Algorithm	Search space	Selected configuration	CV AUROC
CatBoost	iterations ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; depth ∈ {4, 6}	learning_rate = 0.05, iterations = 200, depth = 6	0.8463
XGBoost	n_estimators ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; max_depth ∈ {3, 5}	n_estimators = 200, max_depth = 3, learning_rate = 0.05	0.8446
Gradient Boosting	n_estimators ∈ {100, 200}; learning_rate ∈ {0.01, 0.1}; max_depth ∈ {2, 3}	n_estimators = 200, max_depth = 2, learning_rate = 0.1	0.8443
LightGBM	n_estimators ∈ {200, 500}; learning_rate ∈ {0.01, 0.05, 0.1}; num_leaves ∈ {15, 31}	num_leaves = 15, n_estimators = 500, learning_rate = 0.01	0.8438
SVM	C ∈ {0.1, 1, 10}; kernel ∈ {RBF, linear}	kernel = rbf, C = 1	0.8398
Logistic Regression	C ∈ {0.01, 0.1, 1, 10}; penalty = L2	penalty = l2, C = 0.01	0.8396
Random Forest	n_estimators ∈ {200, 500}; max_depth ∈ {None, 8, 15}; min_samples_leaf ∈ {1, 5}	n_estimators = 500, min_samples_leaf = 5, max_depth = None	0.8377
Decision Tree	max_depth ∈ {3, 5, 8, None}; min_samples_leaf ∈ {1, 5, 20}	min_samples_leaf = 20, max_depth = 5	0.8238
KNN	n_neighbors ∈ {5, 15, 25}	n_neighbors = 25	0.7991
MLP	hidden_layer_sizes ∈ {(64), (64, 32)}; alpha ∈ {0.0001, 0.001}	hidden_layer_sizes = (64), alpha = 0.001	0.7943
Naive Bayes	none (no hyperparameters searched)	—	0.7472

Table S4. Classification performance on the held-out test partition at the decision threshold selected by maximizing the Youden index on out-of-fold training predictions, and at the conventional 0.5 cut-point for comparison. All values except the threshold, F1 and κ are percentages.

Model	Thresh.	Acc.	Sens.	Spec.	PPV	NPV	F1	κ	$Acc._{0.5}$	$Sens._{0.5}$	$Spec._{0.5}$
CatBoost	0.528	77.4	75.3	79.3	75.8	78.8	0.756	0.546	77.5	78.8	76.4
LightGBM	0.510	77.0	77.6	76.5	74.1	79.8	0.758	0.540	76.8	78.5	75.4
XGBoost	0.506	77.3	78.9	75.9	73.9	80.6	0.763	0.546	77.2	79.1	75.5
Gradient Boosting	0.518	77.0	76.6	77.4	74.6	79.3	0.755	0.539	76.9	77.9	76.1
SVM	0.444	75.7	81.7	70.6	70.6	81.7	0.757	0.517	76.4	78.5	74.6
Logistic Regression	0.457	76.7	82.0	72.2	71.8	82.3	0.765	0.536	76.7	77.3	76.2
Random Forest	0.499	76.4	75.2	77.3	74.1	78.3	0.747	0.525	76.4	75.2	77.3
Decision Tree	0.493	74.2	81.0	68.4	68.9	80.7	0.745	0.488	74.2	81.0	68.4
MLP	0.449	72.4	76.9	68.6	67.9	77.4	0.721	0.450	72.6	74.7	70.7
KNN	0.520	72.8	80.9	65.7	67.1	80.0	0.734	0.460	72.8	80.9	65.7
Naive Bayes	0.623	66.3	56.8	74.6	65.9	66.7	0.610	0.317	66.1	61.8	69.8

Acc., accuracy; Sens., sensitivity; Spec., specificity; PPV, positive predictive value; NPV, negative predictive value; κ, Cohen's kappa. Subscript 0.5 denotes the conventional cut-point. Thresholds were selected without reference to the test partition.

Table S5. The fifteen highest-ranked predictors by mean absolute SHAP value, with their permutation-importance values and ranks for comparison. Both methods are applied to the CatBoost model on the held-out test partition and both rank over the same 33 predictors.

Predictor	Mean |SHAP|	SHAP rank	Permutation Δ AUROC (mean ± SD)	Perm. rank	Above noise
Body mass index (kg/m²)	1.0254	1	0.16034 ± 0.00822	1	yes
Age (years)	0.9913	2	0.15632 ± 0.00667	2	yes
Vigorous recreational activity	0.1941	3	0.00191 ± 0.00129	6	yes
Resting pulse (60 sec)	0.1644	4	0.00845 ± 0.00113	3	yes
Race/ethnicity: Non-Hispanic Black	0.0796	5	0.00212 ± 0.00071	5	yes
Self-rated diet quality	0.0747	6	0.00130 ± 0.00076	8	yes
Marital status: Never married	0.0672	7	−0.00047 ± 0.00049	33	no
Education level	0.0630	8	0.00062 ± 0.00044	13	yes
Race/ethnicity: Other/Multi-Racial	0.0462	9	0.00162 ± 0.00037	7	yes
Blood cadmium (µg/L)	0.0437	10	0.00013 ± 0.00058	18	no
Blood lead (µg/dL)	0.0391	11	0.00123 ± 0.00044	9	yes
Blood selenium (µg/L)	0.0388	12	0.00069 ± 0.00023	11	yes
Sex: Female	0.0377	13	0.00254 ± 0.00056	4	yes
Blood mercury, total (µg/L)	0.0362	14	−0.00000 ± 0.00046	27	no
Income-to-poverty ratio	0.0349	15	0.00009 ± 0.00039	19	no

Permutation Δ AUROC is the mean decrease in test-set AUROC over ten permutations of the predictor, with the standard deviation across those permutations. *Above noise* is a descriptive heuristic, not a hypothesis test: it flags predictors whose mean exceeds one permutation standard deviation, a roughly one-sigma bar reflecting permutation variability only and not sampling variability of the test partition. Only 11 of these fifteen clear it, so ranks below the leading few should not be read as a resolved ordering. A negative value means permuting the predictor left discrimination unchanged or marginally improved it. Two of the 33 columns (the missing-category indicators for race/ethnicity and marital status) are constant or near-constant and receive exactly zero importance under both methods; under SHAP they fall to the bottom (joint rank 32), but under permutation importance an exact zero ranks above the seven predictors with negative importance, placing them at joint rank 25 of 33 and pushing genuinely uninformative predictors such as mercury below them.

Table S6. Discrimination and calibration of the CatBoost model within subgroups of the held-out test partition. Calibration is reported alongside discrimination because a model can rank participants poorly within a stratum while still stating their absolute risk correctly, and the two have different consequences in use.

Subgroup	n	Prev. (%)	AUROC (95% CI)	Cal. slope (95% CI)	O:E ratio (95% CI)
Overall	2,229	46.3	0.857 (0.840–0.872)	1.06 (0.97–1.15)	0.970 (0.935–1.000)
Male	1,094	47.6	0.848 (0.824–0.869)	1.05 (0.93–1.20)	0.997 (0.949–1.045)
Female	1,135	45.1	0.865 (0.842–0.885)	1.08 (0.96–1.21)	0.943 (0.900–0.993)
Age 20–40 y	763	23.1	0.861 (0.834–0.888)	1.14 (0.98–1.34)	0.995 (0.894–1.111)
Age 41–60 y	755	45.0	0.826 (0.794–0.855)	1.19 (1.02–1.39)	0.902 (0.845–0.964)
Age > 60 y	711	72.7	0.741 (0.701–0.783)	0.88 (0.71–1.07)	1.011 (0.971–1.052)
Non-Hispanic White	877	50.7	0.850 (0.826–0.874)	1.08 (0.96–1.23)	0.975 (0.924–1.024)
Mexican American	363	45.5	0.864 (0.827–0.900)	1.10 (0.88–1.40)	0.947 (0.864–1.031)
Non-Hispanic Black	478	43.5	0.840 (0.805–0.875)	0.96 (0.80–1.18)	0.955 (0.883–1.035)
Other Hispanic	236	49.6	0.862 (0.816–0.906)	1.16 (0.91–1.53)	0.988 (0.889–1.086)
Other/Multi-racial	275	35.6	0.878 (0.832–0.919)	1.09 (0.84–1.49)	0.995 (0.871–1.127)

Cal. slope, calibration slope, describing dispersion of the predictions (1.00 = perfect); O:E, ratio of observed to expected outcome frequency, describing calibration in the large (1.00 = perfect). All intervals come from the same 1,000 percentile bootstrap resamples, with the calibration model refitted in each. Slope and O:E answer different questions and are not interchangeable. Age groups are tertiles of the test-partition age distribution. Subgroups are nested within the overall sample, so their intervals are not independent of the overall row and should not be used to test against it.

Table S7. Prespecified sensitivity analyses. Each entry is the mean AUROC from 10-fold stratified cross-validation on the training partition, using the three leading algorithms at their default hyperparameter configurations so that the comparison between arms is not confounded by re-tuning. *Absolute values are consequently lower than the tuned values in Table S3 and are not comparable to them; the difference between arms is the quantity of interest.*

Analysis	Model	Primary	Alternative	Δ AUROC
Add diagnosed diabetes	CatBoost	0.8416	0.8515	+0.0099
Add diagnosed diabetes	XGBoost	0.8186	0.8314	+0.0128
Add diagnosed diabetes	Gradient Boosting	0.8442	0.8547	+0.0105
Class weighting instead of SMOTENC	CatBoost	0.8416	0.8420	+0.0005
Class weighting instead of SMOTENC	XGBoost	0.8186	0.8196	+0.0010
Class weighting instead of SMOTENC	Gradient Boosting	0.8442	not applicable	—
Lead only, other four metals dropped	CatBoost	0.8416	0.8427	+0.0011
Lead only, other four metals dropped	XGBoost	0.8186	0.8188	+0.0002
Lead only, other four metals dropped	Gradient Boosting	0.8442	0.8445	+0.0003
Measured-value-only outcome definition	CatBoost	0.8416	0.8097	-0.0319
Measured-value-only outcome definition	XGBoost	0.8186	0.7897	-0.0289
Measured-value-only outcome definition	Gradient Boosting	0.8442	0.8104	-0.0338

Δ AUROC is the alternative specification minus the primary, so a positive value indicates the alternative discriminated better. For the outcome-definition arm the primary specification is the harmonized definition used throughout, and the negative values indicate that the measured-value-only definition is harder to predict.

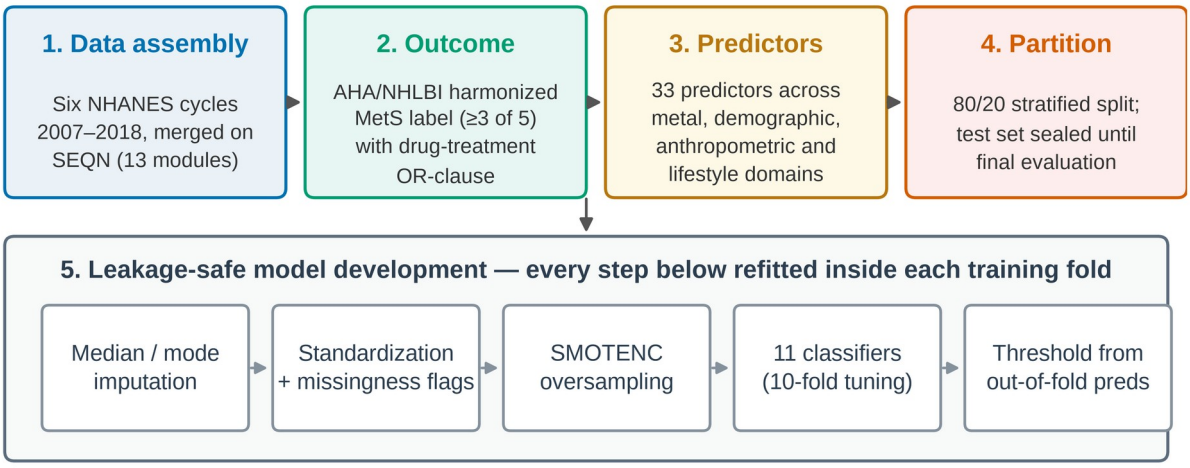


Figure S1. Analysis pipeline. Stages 1–4 are performed once on the cohort. Every step in stage 5 is refitted independently inside each training fold during cross-validation, and inside the training partition alone for the final models, so that no quantity estimated from held-out data can influence a model later scored on it.

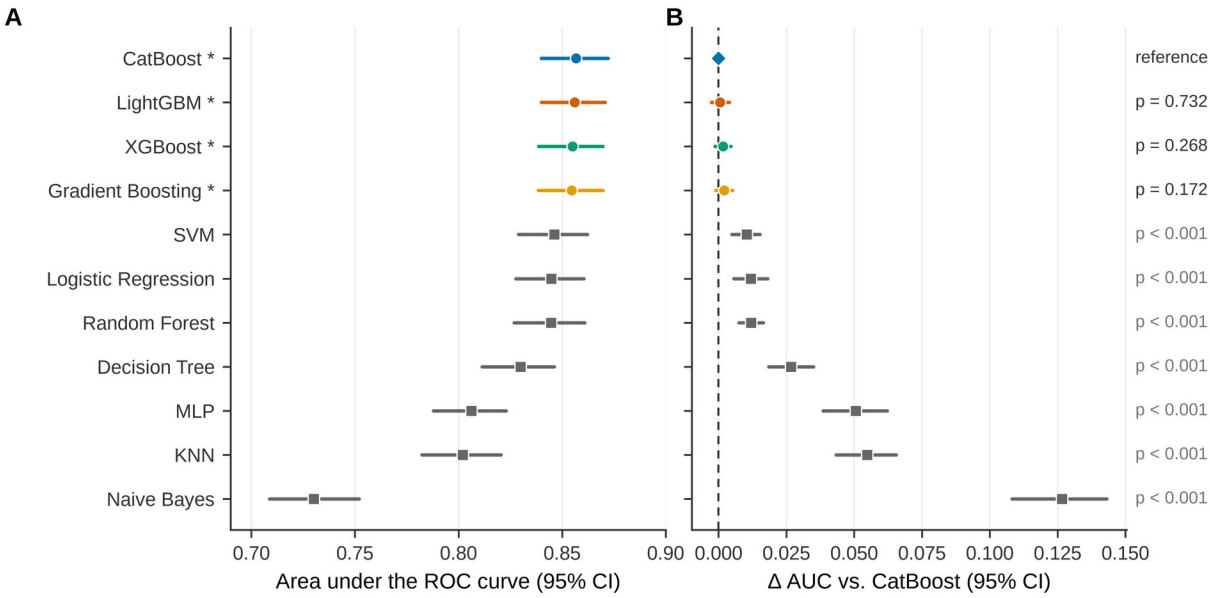


Figure S2. Model discrimination and pairwise comparison against CatBoost. **(A)** AUROC with bootstrap 95% confidence intervals. **(B)** Difference in AUROC from CatBoost from a paired bootstrap in which the same resampled participants were scored under both models; intervals crossing zero indicate models that cannot be distinguished from CatBoost. Using unpaired intervals from panel A for these comparisons would overstate the uncertainty, because all models are scored on the same participants.

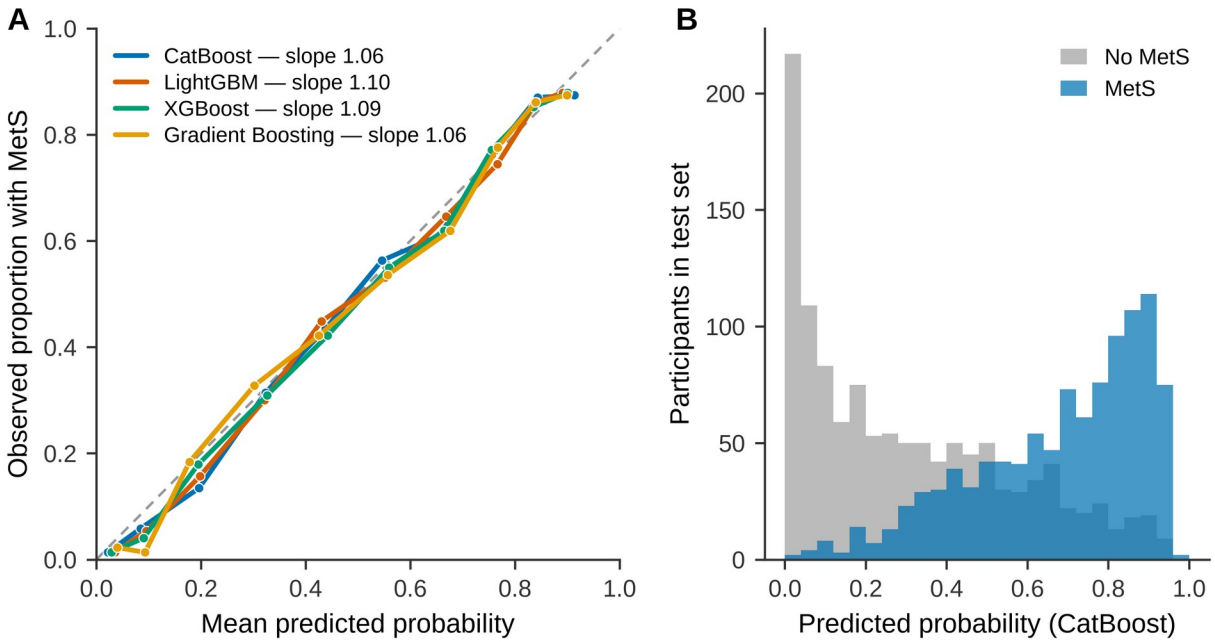


Figure S3. Calibration on the held-out test partition. **(A)** Calibration curves for the four leading models using ten quantile-based bins, with calibration slopes in the legend; the diagonal denotes perfect calibration. **(B)** Distribution of predicted probabilities from CatBoost by true outcome status.

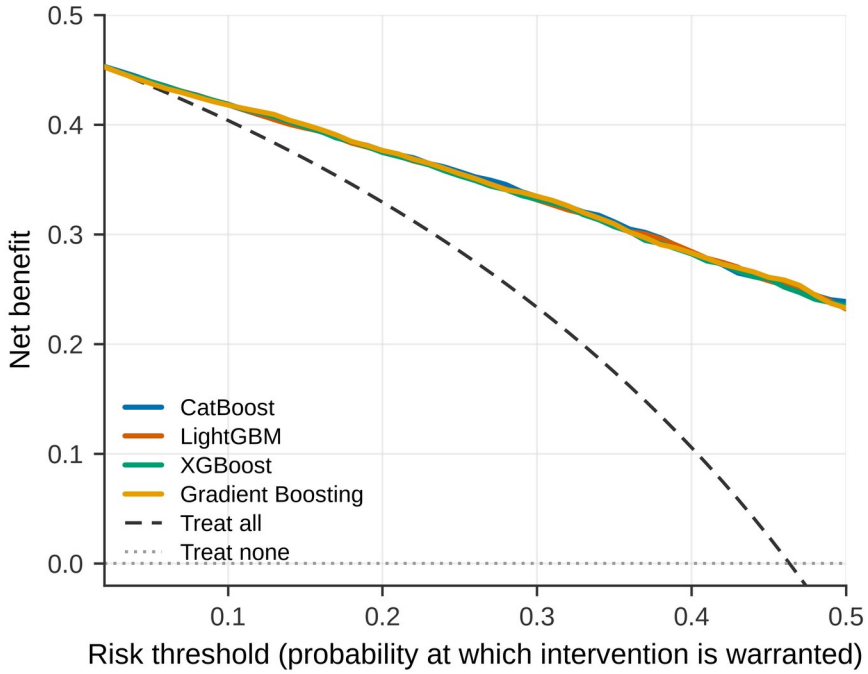


Figure S4. Decision-curve analysis on the held-out test partition. Net benefit is plotted against the risk threshold at which intervention would be considered warranted, for the four leading models and for the treat-all and treat-none reference strategies.

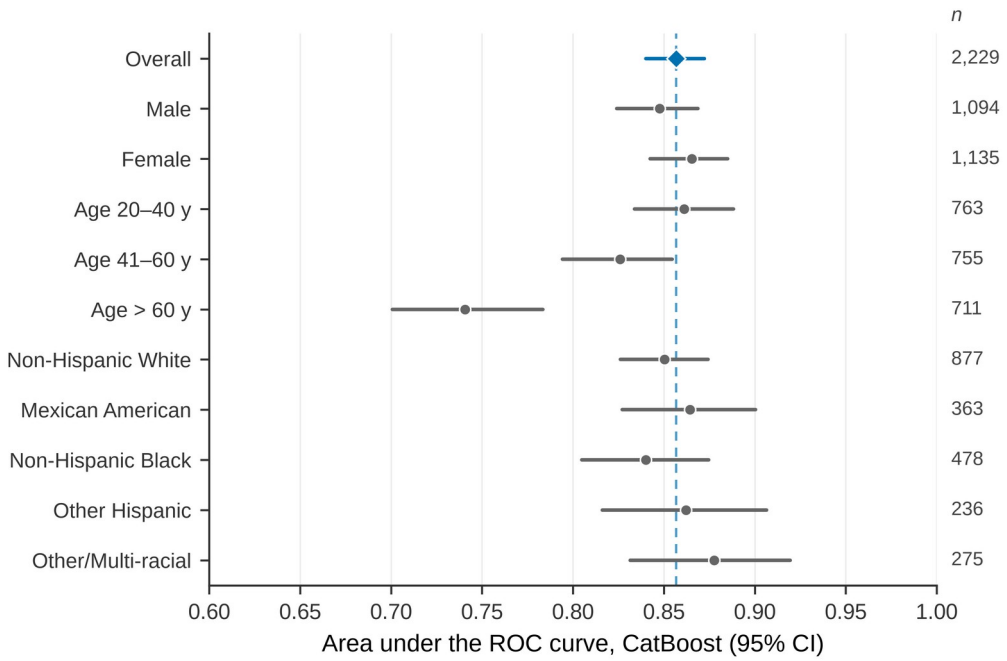


Figure S5. Discrimination of the CatBoost model within subgroups of the held-out test partition, with bootstrap 95% confidence intervals from 1,000 resamples. The dashed vertical line marks the overall test-set AUROC and subgroup sizes are given at the right. Age groups are tertiles of the test-partition age distribution. Because every subgroup is nested within the overall sample, these intervals are not independent of the overall estimate and overlap with it should not be read as a significance test. Subgroup calibration is reported in Table S6.